

HBase / Pig Integration

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HBase / Pig Integration

- *Move big data already in HDFS into HBase using Pig*
- Must create HBase table ahead of time
- *New relation* must have a *unique key* as its first column, followed by subsequent columns
- Both *new relation* and *HBase table* must have exact column
- **USING clause** allows us to STORE data into an HBase table
- Can work at scale – HBase is transactional on rows

very fast

1. Upload data into Ambari [\[HDFS\]](#)

- Login to Ambari
- Upload *u.user* file from movielens dataset into HDFS
- */user/maria_dev/ml-100k/* → **please change this to your own directory**

```
1|24|M|technician|85711
2|53|F|other|94043
3|23|M|writer|32067
4|24|M|technician|43537
5|33|F|other|15213
6|42|M|executive|98101
7|57|M|administrator|91344
8|36|M|administrator|05201
9|29|M|student|01002
10|53|M|lawyer|90703
```

u.user → pipe “|” delimited

Userid
Age
Gender
Occupation
Zipcode

In ambari under files

▼ In puTTY → Create HBase table

```
[ ] # start the HBase interactive shell
hbase shell
```

```
# get the lists of existing tables on HBase instance
list
```

```
# create new table
create 'users','userinfo'
```



Create new table called
“users” with column family
name called “userinfo”

```
# run another list command to check for our newly created table
list
```

```
# exit out from the HBase shell
exit
```

▼ In command prompt [puTTY]

1. Make sure you are in your own directory

2. create a new pig script -> vi hbase.pig

3. Type these following code in the hbase.pig script

- Remember to change the directory accordingly

```
users = LOAD '/user/maria_dev/ml-100k/u.user'
USING PigStorage('|')
AS (userID:int, age:int, gender:chararray, occupation:chararray, zip:int);

STORE users INTO 'hbase://users'
USING org.apache.pig.backend.hadoop.hbase.HBaseStorage (
'userinfo:age,userinfo:gender,userinfo:occupation,userinfo:zip');
```

unique
key

column
family

→ HBase table

→ clause/code use to
transmit data into
HBase

▼ Run the pig script using the following command

```
[ ] pig hbase.pig
```

▼ Return to HBase shell

```
hbase shell

# check whether our "users" table still exists
list

# view what's inside the "users" table
scan 'users'

# Once we are done, we can clean up the mess
disable 'users'
drop 'users'

# now the 'users' table shall no longer exists
list

# exit
exit
```

```
99      column=userinfo:age, timestamp=1483035007446, value=20
99      column=userinfo:gender, timestamp=1483035007446, value=M
99      column=userinfo:occupation, timestamp=1483035007446, value
=student
99      column=userinfo:zip, timestamp=1483035007446, value=63129
```

**timestamp is automatically generated
and appended; “versioning”**